Advanced Causal Inference

Lecture 11: Double/Debiased Machine Learning

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This lecture note is partially based on the lecture notes/textbook of Kennedy (2024); Schuler and van der Laan (2024)

Outline

1. Review
2. Conditions for Ensuring a RAL Estimator
3. Example
4. Assignment

Review

- A principled procedure for constructing an estimator:

1. Specify a causal or statistical model $\mathcal{F}$
2. Identify a functional $\tau(F)$ that characterizes the causal parameter of interest
3. Derive an IF, or if possible, the EIF, for $\tau(F)$ within the model $\mathcal{F}$
4. Construct $\widehat{\tau}$, an estimator for $\tau(F)$ based on the (E)IF
5. Find conditions under which $\widehat{\tau}$ is RAL

- Key properties of the resulting estimator:
  - An IF-based estimator of $\tau(F)$ is robust to bias in the nuisance component estimators i.e., exhibits Neyman orthogonality (Chernozhukov et al., 2018)
  - An EIF-based estimator of $\tau(F)$ achieves the SEB, meaning that
    1. it is the most efficient RAL estimator of $\tau(F)$ in model $\mathcal{F}$; and
    2. further efficiency gains require imposing stronger assumptions on the model

Goal

- Find conditions under which $\widehat{\tau}$ is RAL
- We haven't gone into this topic in detail yet, so we will cover it now

Outline

1 Review

2 Conditions for Ensuring a RAL Estimator

3 Example

4 Assignment

IF-based Estimator

- Suppose that $\varphi(O;\eta) - \tau(F)$ an IF of $\tau(F)$ in a model $\mathcal{F}$ at F^* where η is the nuisance component
  - Example: $\tau(F) = \mathrm{E}_F\{Y(1)\}$. Then,

$$\varphi(O;\eta) - \tau(F) = \frac{A\{Y - \mu(1,X)\}}{e(X)} + \mu(1,X) - \tau(F)\ , \quad \eta = (\mu, e)$$

- We let $\tau^* = \tau(F^*)$ and $\eta^* = \eta(F^*)$

- An IF-based estimator is

$$\widehat{\tau} = \mathbb{P}_{\mathcal{I}}\big[\varphi(O;\widehat{\eta})\big]$$

where

$$\widehat{\eta} = \text{a generic estimator of } \eta; \qquad \mathbb{P}_{\mathcal{I}}\big[f(O)\big] = \frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} f(O_i)$$

- Note: the "oracle" estimator $\widetilde{\tau} = \mathbb{P}_{\mathcal{I}}\big[\varphi(O;F^*)\big]$ where the true distribution F^* is used

IF-based Estimator

- The oracle estimator is RAL, and thus, AN from the CLT:

$$\begin{aligned}\sqrt{N}(\tilde{\tau} - \tau^*) &= \sqrt{N}\left[\frac{1}{N}\sum_{i=1}^{N}\varphi(O_i;\eta^*) - \tau^*\right] \\ &= \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\{\varphi(O_i;\eta^*) - \tau^*\} \\ &\xrightarrow{D} N\Big(0, \mathrm{Var}\{\varphi(O;\eta^*)\}\Big)\end{aligned}$$

- Of course, this estimator is infeasible to obtain since the estimator requires knowledge of the true distribution

IF-based Estimator

- The feasible estimator's $\sqrt{N}$-factored error looks like:

$$
\begin{aligned}
&\sqrt{N}(\hat{\tau} - \tau^*) \\
&= \sqrt{N}\Big[\frac{1}{N}\sum_{i=1}^{N}\varphi(O_i;\hat{\eta}) - \tau^*\Big] \\
&= \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\{\varphi(O_i;\hat{\eta}) - \tau^*\} \\
&= \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\{\varphi(O_i;\eta^*) - \tau^*\} + \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\{\varphi(O_i;\hat{\eta}) - \varphi(O_i;\eta^*)\} \\
&= \underbrace{\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\{\varphi(O_i;\eta^*) - \tau^*\}}_{\text{Term 1}} + \underbrace{\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\begin{bmatrix}[\varphi(O_i;\hat{\eta}) - \mathrm{E}_{O'\sim F^*}\{\varphi(O';\hat{\eta})\}] \\ -[\varphi(O_i;\eta^*) - \mathrm{E}_{O'\sim F^*}\{\varphi(O';\eta^*)\}]\end{bmatrix}}_{\text{Term 2}} + \underbrace{\sqrt{N}\begin{bmatrix}\mathrm{E}_{O'\sim F^*}\{\varphi(O';\hat{\eta})\} \\ -\mathrm{E}_{O'\sim F^*}\{\varphi(O';\eta^*)\}\end{bmatrix}}_{\text{Term 3}}
\end{aligned}
$$

IF-based Estimator

$$\sqrt{N}(\hat{\tau}-\tau^*) = \underbrace{\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\{\varphi(O_i;\eta^*)-\tau^*\}}_{\text{Term 1}} + \underbrace{\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\begin{bmatrix}[\varphi(O_i;\hat{\eta})-\mathrm{E}_{O'\sim F^*}\{\varphi(O';\hat{\eta})\}] \\ -[\varphi(O_i;\eta^*)-\mathrm{E}_{O'\sim F^*}\{\varphi(O';\eta^*)\}]\end{bmatrix}}_{\text{Term 2}} + \underbrace{\sqrt{N}\begin{bmatrix}\mathrm{E}_{O'\sim F^*}\{\varphi(O';\hat{\eta})\} \\ -\mathrm{E}_{O'\sim F^*}\{\varphi(O';\eta^*)\}\end{bmatrix}}_{\text{Term 3}}$$

- Term 1 is the CLT part:

$$\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\{\varphi(O_i;\eta^*)-\tau^*\} \xrightarrow{D} N\big(0, \mathrm{Var}\{\varphi(O_i;\eta^*)\}\big)$$

- Therefore, if Term 2 and Term 3 are $o_P(1)$, then $\hat{\tau}$ is RAL for τ^*

IF-based Estimator

- Term 2 is the empirical process:

$$\mathbb{G}_{\mathcal{I}}\{\varphi(O;\widehat{\eta}) - \varphi(O;\eta^*)\} = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \begin{bmatrix} \left[\varphi(O_i;\widehat{\eta}) - \mathrm{E}_{O' \sim F^*}\{\varphi(O';\widehat{\eta})\}\right] \\ -\left[\varphi(O_i;\eta^*) - \mathrm{E}_{O' \sim F^*}\{\varphi(O';\eta^*)\}\right] \end{bmatrix}$$

- Term 3 is the bias from the nuisance component estimation

$$\text{Bias}_{\varphi}(\widehat{\eta},\eta^*) = \sqrt{N} \begin{bmatrix} \mathrm{E}_{O' \sim F^*}\{\varphi(O';\widehat{\eta})\} \\ -\mathrm{E}_{O' \sim F^*}\{\varphi(O';\eta^*)\} \end{bmatrix}$$

- **Goal**: find conditions under which Term 2 and Term 3 are $o_P(1)$

Empirical Process Term

$$\text{Term 2} = \mathbb{G}_{\mathcal{I}}\{\varphi(O;\widehat{\eta}) - \varphi(O;\eta^*)\} = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \begin{bmatrix} [\varphi(O_i;\widehat{\eta}) - \mathrm{E}_{O' \sim F^*}\{\varphi(O';\widehat{\eta})\}] \\ -[\varphi(O_i;\eta^*) - \mathrm{E}_{O' \sim F^*}\{\varphi(O';\eta^*)\}] \end{bmatrix}$$

- In order to ensure $\mathbb{G}_{\mathcal{I}}\{\varphi(O;\widehat{\eta}) - \varphi(O;\eta^*)\} = o_P(1)$, we need:

Condition 1: Consistency

$$\|\varphi(O;\widehat{\eta}) - \varphi(O;\eta^*)\|_{P,2}^2 = \int \{\varphi(O;\widehat{\eta}) - \varphi(O;\eta^*)\}^2 \, dF(O) = o_P(1)$$

and one of:

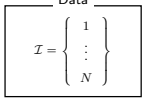
Condition 2-1: Use sample splitting & cross-fitting or

Condition 2-2: Estimate η such that $\varphi(O;\eta)$ is in a Donsker class

- Condition 1 simply requires that the estimator $\widehat{\eta}$ converges to the true η^*.
  - Many ML methods satisfy this condition!
- Condition 2 is required to address the double-dipping issue
  - The same data is used twice: once to estimate $\widehat{\eta}$ and again in the average $N^{-1}\sum_{i=1}^{N}$
  - If $\widehat{\eta}$ is obtained via a ML method, it becomes difficult to analytically characterize Term 2

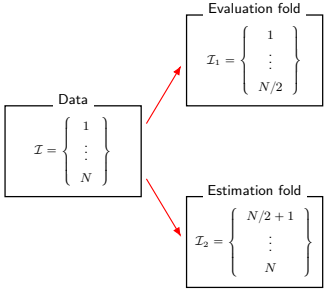
Controlling the Empirical Process Term: Sample Splitting

- Schick (1986); Chernozhukov et al. (2018)



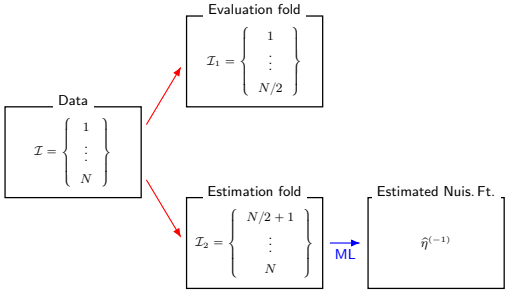
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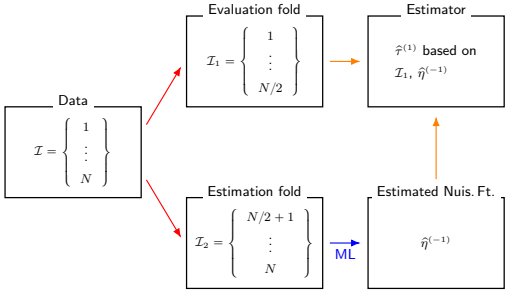
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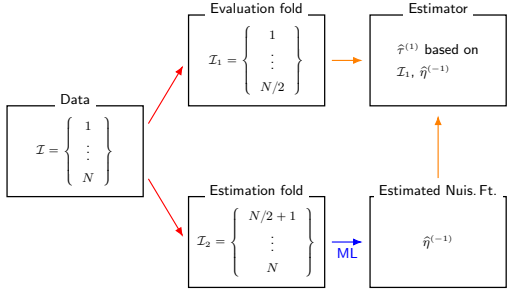
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Controlling the Empirical Process Term: Sample Splitting

- Schick (1986); Chernozhukov et al. (2018)



- The estimator $\widehat{\tau}^{(1)}$ has an emp. proc. term:

$$[\text{Term 2}](\mathcal{I}_1) = \frac{1}{\sqrt{N/2}} \sum_{i \in \mathcal{I}_1} \left[\begin{array}{l} \left[\varphi(O_i; \widehat{\eta}^{(-1)}) - \mathrm{E}_{O' \sim F^*}\{\varphi(O'; \widehat{\eta}^{(-1)})\}\right] \\ -\left[\varphi(O_i; \eta^*) - \mathrm{E}_{O' \sim F^*}\{\varphi(O'; \eta^*)\}\right] \end{array} \right]$$

- By viewing $\widehat{\eta}^{(-1)}$ as a "fixed" function, one can show that $[\text{Term 2}](\mathcal{I}_1) = o_P(1)$
- The estimator only use $N/2$ observations, which can result in a substantial loss of efficiency

Controlling the Empirical Process Term: Donsker Class

- A Donsker class is just a family of functions that are not "too complicated"; examples include:
  - finite collection of functions
  - VC classes of bounded functions
  - Lipschitz functions on a compact domain
  - Sobolev classes with sufficient smoothness
  - Monotone functions

- The following lemma guarantees Term 2 is $o_P(1)$ if $\varphi(O;\eta)$ is in a Donsker class:

Lemma (Lemma 19.24 of van der Vaart (1998))

Suppose that $\mathcal{F}$ is a P-Donsker class of measurable functions and $\widehat{f}_n$ is a sequence of random functions that take their values in $\mathcal{F}$ such that $\int(\widehat{f}_n(x) - f_0(x))^2\, dP(x)$ converges in probability to 0 for some $f_0 \in L_2(P)$. Then, $\mathbb{G}_n(\widehat{f}_n - f_0) = o_P(1)$.

- However, it remains an open question whether the function spaces induced by ML methods (e.g., deep neural networks, random forests, boosted trees, etc.) are Donsker

- Although useful, Lemma 19.24 may not be applicable to modern ML methods

Bias Term

$$\text{Term 3} = \text{Bias}_\varphi(\widehat{\eta}, \eta^*) = \sqrt{N}\begin{bmatrix} \mathrm{E}_{O' \sim F^*}\{\varphi(O'; \widehat{\eta})\} \\ -\mathrm{E}_{O' \sim F^*}\{\varphi(O'; \eta^*)\} \end{bmatrix}$$

- Recall the von Mises representation (distributional Taylor expansion) gives

$$\tau(G) - \tau(F)\beta(G) - \beta(F) = \int \varphi_G(z)\, d(G-F)(z) + R_2(G, F) = \int \varphi_G(z)\, d(G-F)(z) + O\big(\|G-F\|^2\big)$$

- It is important to note that R_2 is at most of second order with respect to the change in the distribution

- Therefore, the bias term can be written as:

$$\begin{aligned}
&\mathrm{E}_{O' \sim F^*}\{\varphi(O'; \widehat{\eta})\} - \mathrm{E}_{O' \sim F^*}\{\varphi(O'; \eta^*)\} \\
&= \mathrm{E}_{O' \sim F_{\widehat{\eta}}}\{\varphi(O'; \widehat{\eta})\} - \mathrm{E}_{O' \sim F_{\eta^*}}\{\varphi(O'; \eta^*)\} - \mathrm{E}_{O' \sim F_{\widehat{\eta}}}\{\varphi(O'; \widehat{\eta})\} + \mathrm{E}_{O' \sim F_{\eta^*}}\{\varphi(O'; \widehat{\eta})\} \\
&= \tau(F_{\widehat{\eta}}) - \tau^* - \int \varphi(O'; \widehat{\eta})\, d(F_{\widehat{\eta}} - F_{\eta^*})(O) \\
&= R_2(F_{\widehat{\eta}}, F_{\eta^*}) \\
&= O\big(\|\widehat{\eta} - \eta^*\|^2\big)
\end{aligned}$$

Bias Term

- Therefore, what we have established:

$$\text{Term 3} = \text{Bias}_{\varphi}(\widehat{\eta}, \eta^*) = \sqrt{N}\begin{bmatrix} \mathrm{E}_{O' \sim F^*}\{\varphi(O'; \widehat{\eta})\} \\ -\mathrm{E}_{O' \sim F^*}\{\varphi(O'; \eta^*)\} \end{bmatrix} = \sqrt{N} \times O\Big(\|\widehat{\eta} - \eta\|^2\Big)$$

- Therefore, if $\|\widehat{\eta} - \eta^*\|_{P,2} = o_P(N^{-1/4})$, we have [Term 3] $= o_P(1)$

- This rate is attainable provided that:
  - The true nuisance function η^* is not overly complex (e.g., sufficiently smooth); and
  - The estimator can adapt to this level of complexity

- For example, $\eta = \eta(X)$ with $\dim(X) = d$
  If $\eta(X)$ is s-Hölder smooth, the local polynomial estimator satisfies $\|\widehat{\eta} - \eta^*\|_{P,2} = O_P(N^{-s/(2s+d)})$
  Therefore, when $s > d/2$, then $o_P(N^{-1/4})$ rate is attainable

Neyman Orthogonality

- The fact that the bias is of second order is because we used an IF to construct an estimator
- In other words, an IF-based estimator inherently exhibits at most second-order bias
  - This is a property commonly referred to as Neyman orthogonality (Chernozhukov et al., 2018)
- Conversely, if the estimator is not IF-based, such as a plug-in estimator, it may exhibit first-order bias with respect to the nuisance function η
- Specifically, [Term 3] $= \sqrt{N} \times O(\|\hat{\eta} - \eta\|)$
- Since nonparametric estimators typically converge substantially slower than $\sqrt{N}$, this implies that [Term 3] is generally **NOT** $o_P(1)$; in other words, the bias can be substantial

Mixed-bias Property

- Suppose the nuisance component η can be decomposed into η_1 and η_2
  - Example: $\tau(F) = \mathrm{E}_F\{Y(1)\}$. Then,

$$\varphi(O;\eta) - \tau(F) = \frac{A\{Y - \mu(1,X)\}}{e(X)} + \mu(1,X) - \tau(F)\ , \quad \eta = (\mu, e) \tag{1}$$

- Since the bias of an IF-based estimator is at most of second order, it can be expressed as

$$\text{Bias} = O\Big(\|\hat{\eta} - \eta^*\|^2\Big) = O\Big(\|\hat{\eta}_1 - \eta_1^*\|^2\Big) + O\Big(\|\hat{\eta}_2 - \eta_2^*\|^2\Big) + O\Big(\|\hat{\eta}_1 - \eta_1^*\|\|\hat{\eta}_2 - \eta_2^*\|\Big)$$

- Mixed-bias property (Rotnitzky et al., 2020):
  the bias structure contains no quadratic term in any single component, but only cross-product terms:

$$\text{Bias} = O\Big(\|\hat{\eta}_1 - \eta_1^*\|\|\hat{\eta}_2 - \eta_2^*\|\Big)$$

- The mixed-bias property is the fundamental reason why an estimator exhibits the double robustness property
  - The AIPW esimator based on (1) is doubly robust because it has the mixed-bias property w.r.t. e and μ
- Neyman orthogonality (bias = second order) is weaker than the mixed-bias property
  - Therefore, although all influence functions exhibit Neyman orthogonality, not all exhibit the mixed-bias property

Example: Counterfactual mean

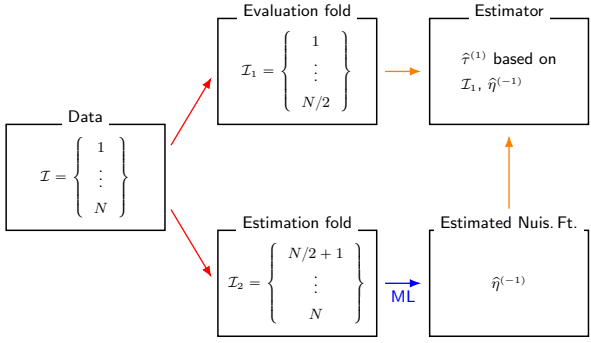
- The bias term of the IF-based estimator for $\mathrm{E}_F\{\mathrm{E}_F(Y|A=1,X)\}$ is

$$
\begin{aligned}
&\sqrt{N} \times \left[\mathrm{E}_{O' \sim F^*}\{\varphi(O';\widehat{\eta})\} - \mathrm{E}_{O' \sim F^*}\{\varphi(O';\eta^*)\}\right] \\
&= \sqrt{N} \times \mathrm{E}_{O'}\left[\frac{A'\{Y' - \widehat{\mu}(1,X')\}}{\widehat{e}(X')} + \widehat{\mu}(1,X') - \mu^*(1,X')\right] \\
&= \sqrt{N} \times \mathrm{E}_{O'}\left[\frac{e^*(X')\{\mu^*(1,X') - \widehat{\mu}(1,X')\}}{\widehat{e}(X')} + \widehat{\mu}(1,X') - \mu^*(1,X')\right] \\
&= \sqrt{N} \times \mathrm{E}_{O'}\left[\frac{\{e^*(X') - \widehat{e}(X')\}\{\mu^*(1,X') - \widehat{\mu}(1,X')\}}{\widehat{e}(X')}\right] \\
&\leqslant c^{-1} \times \sqrt{N} \times \mathrm{E}_{O'}\Big[\{e^*(X') - \widehat{e}(X')\}\{\mu^*(1,X') - \widehat{\mu}(1,X')\}\Big] && \text{Assuming that } \widehat{e}(x) \in [c, 1-c] \\
&\leqslant c^{-1} \times \sqrt{N} \times \|e^*(X') - \widehat{e}(X')\|_{P,2} \|\mu^*(1,X') - \widehat{\mu}(1,X')\|_{P,2} && \text{Cauchy-Schwartz}
\end{aligned}
$$

- Therefore, we only need $\|e^*(X') - \widehat{e}(X')\|_{P,2} \|\mu^*(1,X') - \widehat{\mu}(1,X')\|_{P,2} = o_P(N^{-1/2})$
- One component may have a larger error than the other, provided that the cross-product error is $o_P(N^{-1/2})$
  - If e is known (i.e., randomized experiment), it suffices to assume only that $\|\mu^* - \widehat{\mu}\|_{P,2} = o_P(1)$; converge at **ANY** rate
  - Of course, the condition is satisfied when both components' error are $o_P(N^{-1/4})$

Double/Debiased Machine Learning

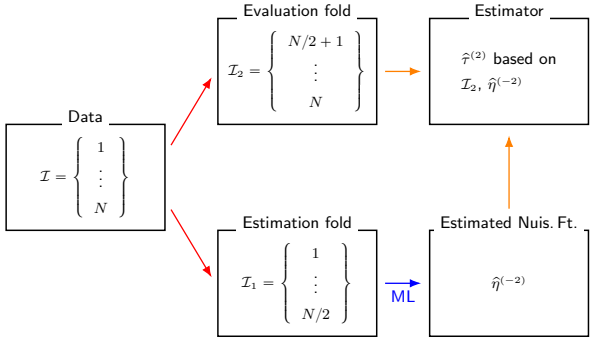
- Chernozhukov et al. (2018): DDML



- Sample splitting with $(\mathcal{I}_1, \mathcal{I}_2)$

Double/Debiased Machine Learning

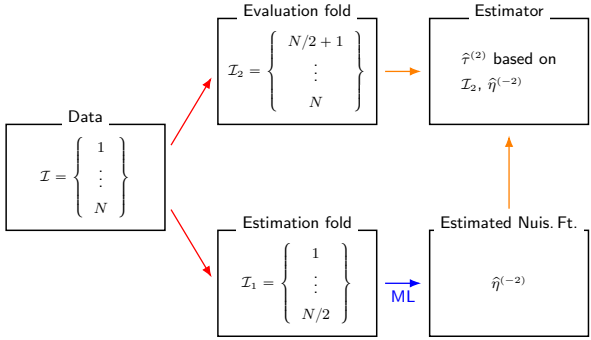
- Chernozhukov et al. (2018): DDML



- Sample splitting with $(\mathcal{I}_1, \mathcal{I}_2)$ and with $(\mathcal{I}_2, \mathcal{I}_1)$

Double/Debiased Machine Learning

- Chernozhukov et al. (2018): DDML



- Sample splitting with $(\mathcal{I}_1, \mathcal{I}_2)$ and with $(\mathcal{I}_2, \mathcal{I}_1)$
- Cross-fitting: by averaging the two resulting estimators, we can regain full efficiency

$$\widehat{\tau}_{\mathsf{CF}} = \frac{1}{N} \sum_{k=1}^{2} \sum_{i \in \mathcal{I}_k} \varphi(O_i; \widehat{\eta}^{(-k)})$$

Double/Debiased Machine Learning

- The cross-fitting estimator is RAL when nuisance functions are estimated at $o_P(N^{-1/4})$ rate

$$\widehat{\tau}_{\mathsf{CF}} = \frac{1}{N}\sum_{k=1}^{2}\sum_{i\in\mathcal{I}_k}\varphi(O_i;\widehat{\eta}^{(-k)}) \xrightarrow{D} N\big(0,\sigma^2 := \mathrm{Var}\{\varphi(O;\eta^*)\}\big) \tag{2}$$

- Again, $\widehat{\tau}_{\mathsf{CF}}$ is RAL because it's built directly from an IF
- A consistent variance estimator is the sample variance:

$$\widehat{\sigma}^2_{\mathsf{CF}} = \frac{1}{N}\sum_{k=1}^{2}\sum_{i\in\mathcal{I}_k}\Big\{\varphi(O_i;\widehat{\eta}^{(-k)}) - \widehat{\tau}_{\mathsf{CF}}\Big\}^2 \tag{3}$$

- A 95% CI:

$$\left(\widehat{\tau}_{\mathsf{CF}} - 1.96\frac{\widehat{\sigma}_{\mathsf{CF}}}{\sqrt{N}} \quad , \quad \widehat{\tau}_{\mathsf{CF}} + 1.96\frac{\widehat{\sigma}_{\mathsf{CF}}}{\sqrt{N}}\right)$$

- While the asymptotic distribution does not depend on the specific realization of split samples, finite-sample performance can be heavily influenced by it

- Chernozhukov et al. (2018) proposed median adjustment

Algorithm 1: Median Adjustment

for $s = 1, \ldots, S$ **do**

Randomly split data into $\mathcal{I}_1^{[s]}$ and $\mathcal{I}_2^{[s]}$;

Obtain a CF estimate $\hat{\tau}_{\mathsf{CF}}^{[s]}$ in (2), and the corresponding variance estimate $\hat{\sigma}_{\mathsf{CF}}^{2,[s]}$ in (3)

end

Consider the median-adjusted CF estimate and its variance estimate

$$\hat{\tau}_{\mathsf{CF}}^{\mathsf{med}} = \underset{s=1,\ldots,S}{\mathrm{median}}\, \hat{\tau}_{\mathsf{CF}}^{[s]} \ , \qquad \hat{\sigma}_{\mathsf{CF}}^{2,\mathsf{med}} = \underset{s=1,\ldots,S}{\mathrm{median}} \left[\hat{\sigma}_{\mathsf{CF}}^{2,[s]} + \left(\hat{\tau}_{\mathsf{CF}}^{[s]} - \hat{\tau}_{\mathsf{CF}}^{\mathsf{med}}\right)^2\right]$$

return $\hat{\tau}_{\mathsf{CF}}^{\mathsf{med}}$ and $\hat{\sigma}_{\mathsf{CF}}^{2,\mathsf{med}}$

- A 95% CI:

$$\left(\hat{\tau}_{\mathsf{CF}}^{\mathsf{med}} - 1.96\frac{\hat{\sigma}_{\mathsf{CF}}^{\mathsf{med}}}{\sqrt{N}} \quad , \quad \hat{\tau}_{\mathsf{CF}}^{\mathsf{med}} + 1.96\frac{\hat{\sigma}_{\mathsf{CF}}^{\mathsf{med}}}{\sqrt{N}}\right)$$

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4 Assignment

Example: NHEFS

- `causaldata::nhefs_complete`:

  National Health and Nutrition Examination Survey Epidemiologic Follow-up Study

Role	Variable	Description	Example Values / Notes
X	sex	Sex	0 = Female, 1 = Male
	race	Race	1 = White, 2 = Black, 3 = Other
	age	Age at baseline	e.g., 25–74
	school	Years of schooling	e.g., 0–20
	smokeintensity	Number of cigarettes smoked per day in 1971	e.g., 0–60
	smokeyrs	Years of smoking	e.g., 0–50
	exercise	Recreational exercise in 1971	0 = much, 1 = moderate, 2 = little/no
	active	Usual daily activity in 1971	0 = very active, 1 = moderately, 2 = inactive, 3 = missing
	wt71	Weight in kilograms in 1971	e.g., 45–120
A	qsmk	Quit smoking between 1st questionnaire and 1982	0 = No, 1 = Yes
Y	wt82	Weight in kilograms in 1982	e.g., 45–130
	wt82_71	Weight change (wt82 - wt71)	Can be negative or positive

- Study the effect of quit smoking on weight in 1982

Example: NHEFS

```
library(causaldata); library(randomForest)

data <- nhefs_complete; N <- nrow(data)

Est <- rep(0,10); SD <- rep(0,10)
for(ss in 1:10){
  SS.Index <- list();  SS.Index[[1]] <- sample(N,round(N/2)); SS.Index[[2]] <- (1:N)[-SS.Index[[1]]]

  PS.Fit <- OR.Fit <- list()
  for(tt in 1:2){
    PS.Fit[[tt]] <- randomForest(factor(qsmk)~sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71,  data=data[SS.Index[[tt]],])
    OR.Fit[[tt]] <- randomForest(wt82~qsmk+sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71,  data=data[SS.Index[[tt]],])
  }

  data1 <- data0 <- data; data1$qsmk <- 1; data0$qsmk <- 0
  PS.Est <- OR1.Est <- OR0.Est <- rep(0,nrow(data))
  for(tt in 1:2){
    PS.Est[SS.Index[[3-tt]]] <- predict(PS.Fit[[tt]],type="prob",newdata=data[SS.Index[[3-tt]],])[,"1"]
    OR1.Est[SS.Index[[3-tt]]] <- predict(OR.Fit[[tt]],newdata=data1[SS.Index[[3-tt]],])
    OR0.Est[SS.Index[[3-tt]]] <- predict(OR.Fit[[tt]],newdata=data0[SS.Index[[3-tt]],])
  }

  Y <- data$wt82; A <- data$qsmk
  ATE.IF.CF <- A*(Y-OR1.Est)/PS.Est-(1-A)*(Y-OR0.Est)/(1-PS.Est)+(OR1.Est-OR0.Est)
  Est[ss] <- mean(ATE.IF.CF); SD[ss] <- sd(ATE.IF.CF)/sqrt(N)
}
```

Example: NHEFS

- One can use `SuperLearner` package to implement ensemble ML methods

```
library(SuperLearner)
data_X <- data[,c("sex","race","age","school","smokeintensity","smokeyrs","exercise","active","wt71")]
data_AX <- data[,c("qsmk","sex","race","age","school","smokeintensity","smokeyrs","exercise","active","wt71")]
data_AX1 <- data_AX0 <- data_AX; data_AX1$qsmk <- 1; data_AX0$qsmk <- 0
SL.library <- c("SL.glmnet", "SL.glm", "SL.randomForest", "SL.gam", "SL.mean")

Est <- rep(0,5); SD <- rep(0,5)
for(ss in 1:5){
  SS.Index <- list(); SS.Index[[1]] <- sample(N,round(N/2)); SS.Index[[2]] <- (1:N)[-SS.Index[[1]]]
  Y <- data$wt82; A <- data$qsmk

  PS.Fit <- OR.Fit <- list()
  for(tt in 1:2){
    PS.Fit[[tt]] <- SuperLearner(Y=A[SS.Index[[tt]]],X=data_X[SS.Index[[tt]],],SL.library = SL.library,family=binomial())
    OR.Fit[[tt]] <- SuperLearner(Y=Y[SS.Index[[tt]]],X=data_AX[SS.Index[[tt]],],SL.library = SL.library,family=gaussian())
  }

  data1 <- data0 <- data; data1$qsmk <- 1; data0$qsmk <- 0
  PS.Est <- OR1.Est <- OR0.Est <- rep(0,nrow(data))
  for(tt in 1:2){
    PS.Est[SS.Index[[3-tt]]]  <- predict(PS.Fit[[tt]],data_X[SS.Index[[3-tt]],])$pred
    OR1.Est[SS.Index[[3-tt]]] <- predict(OR.Fit[[tt]],data_AX1[SS.Index[[3-tt]],])$pred
    OR0.Est[SS.Index[[3-tt]]] <- predict(OR.Fit[[tt]],data_AX0[SS.Index[[3-tt]],])$pred
  }

  ATE.IF.CF <- A*(Y-OR1.Est)/PS.Est-(1-A)*(Y-OR0.Est)/(1-PS.Est)+(OR1.Est-OR0.Est)
  Est[ss] <- mean(ATE.IF.CF); SD[ss] <- sd(ATE.IF.CF)/sqrt(N)
}
```

Example: NHEFS

```
Est.CF.Median <- median(Est)
SD.CF.Median <- median( SD^2 + (Est-Est.CF.Median)^2 )^(0.5)

round( c(Est.CF.Median, SD.CF.Median,
         Est.CF.Median -1.96*SD.CF.Median,
         Est.CF.Median +1.96*SD.CF.Median), 3)
```

- $\widehat{\tau}^{\text{med}}_{\text{CF, RF}} = 3.649$, SE $= 0.746$, 95% CI $= (2.228, 5.151)$
- $\widehat{\tau}^{\text{med}}_{\text{CF, SL}} = 3.355$, SE $= 0.476$, 95% CI $= (2.422, 4.288)$
- Parametric model-based:
  - $\widehat{\tau}_{\text{TMLE}} = 3.329$, SE $= 0.246$, 95% CI $= (2.357, 4.301)$
  - $\widehat{\tau}_{\text{AIPW}} = 3.313$, SE $= 0.489$, 95% CI $= (2.393, 4.280)$

Example: NHEFS

- One can also use `DoubleML` packages
- See examples at https://thomaswiemann.com/ddml/

Outline

1 Review

2 Conditions for Ensuring a RAL Estimator

3 Example

4 Assignment

Assignment

- Analyze the dataset `causalweight::JC` (Job Corps)
  While the dataset contains additional variables, we will focus only on the following:
  - $X =$ (`female, age, white, black, hispanic`)
  - $A =$ `assignment`
  - $Y =$ `earny4`

  We aim to study the effect of Job Corps program assignment on weekly earnings in the fourth year after assignment

- Estimate the ATE, and construct a 95% CI using:
  - A cross-fitted estimator without median adjustment
  - A median-adjusted cross-fitted estimator (10 repetitions)

- For both estimators, implement them in two ways:
  - With a self-coded step (similar to the example code)
  - Using the `DoubleML` package

- You may use your favorite ML methods to estimated nuisance functions

- Be sure to interpret the point estimate and inference results, as understanding how to draw the causal conclusion is the key focus of this course

References

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